

Drug Delivery: Engineering and Principles

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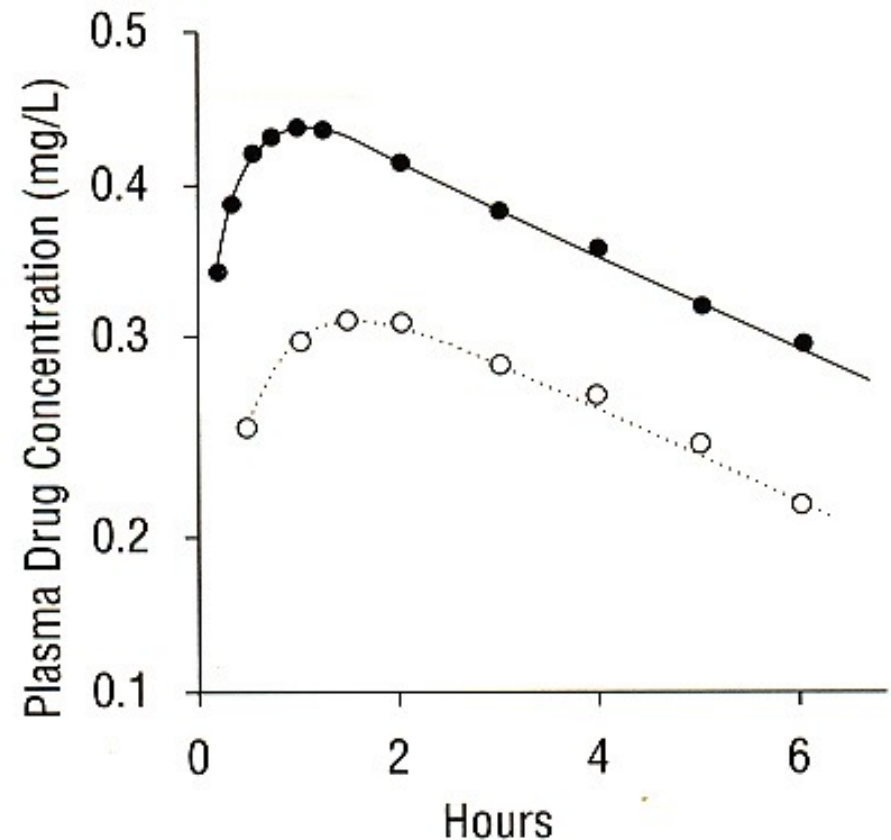
Indian Institute of Science

Route specific delivery

- Depending upon application appropriate routes and specific delivery systems are chosen

- 500 mg dose given
 - intramuscularly
 - orally

**to the same subject on separate occasions



Oral Administration

- Advantages
 - Patient: Convenience, not invasive, higher compliance
 - Manufacture: well established processes, available infrastructure
- Disadvantages
 - Unconscious patients cannot take dose
 - Low solubility
 - Low permeability
 - Degradation by GI enzymes or flora
 - First pass metabolism
 - Food interactions
 - Irregular absorption

Oral Administration

- Traditional oral delivery systems
 - Tablets
 - Capsules
 - Soft gelatin capsules
 - Suspensions

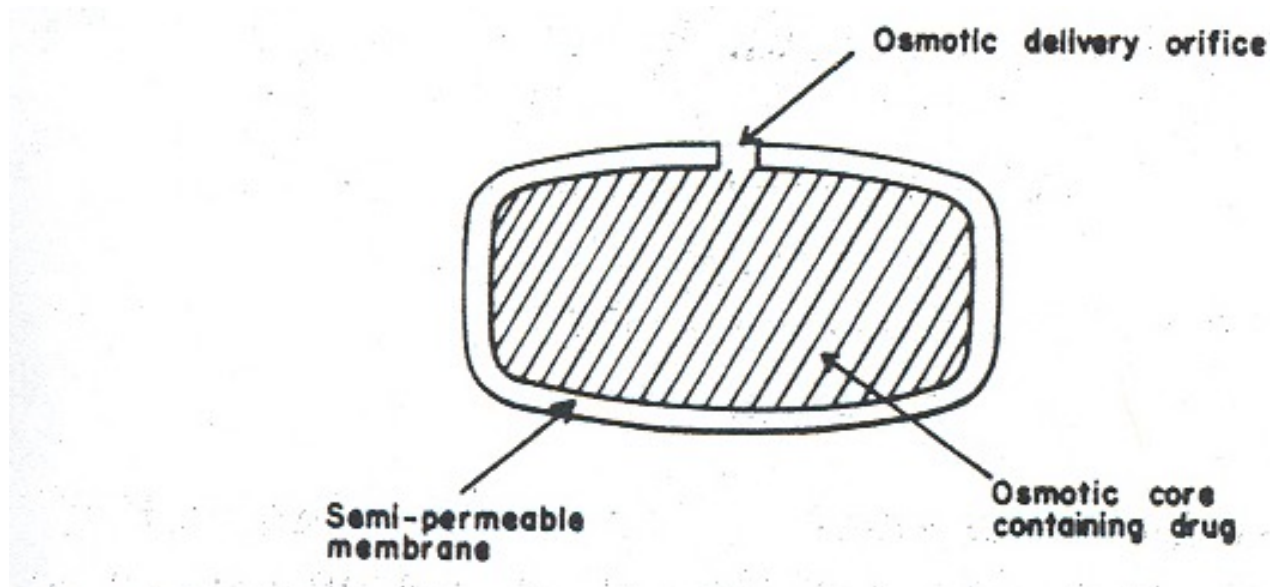


Controlled release dosage forms

Generally for highly water soluble drugs

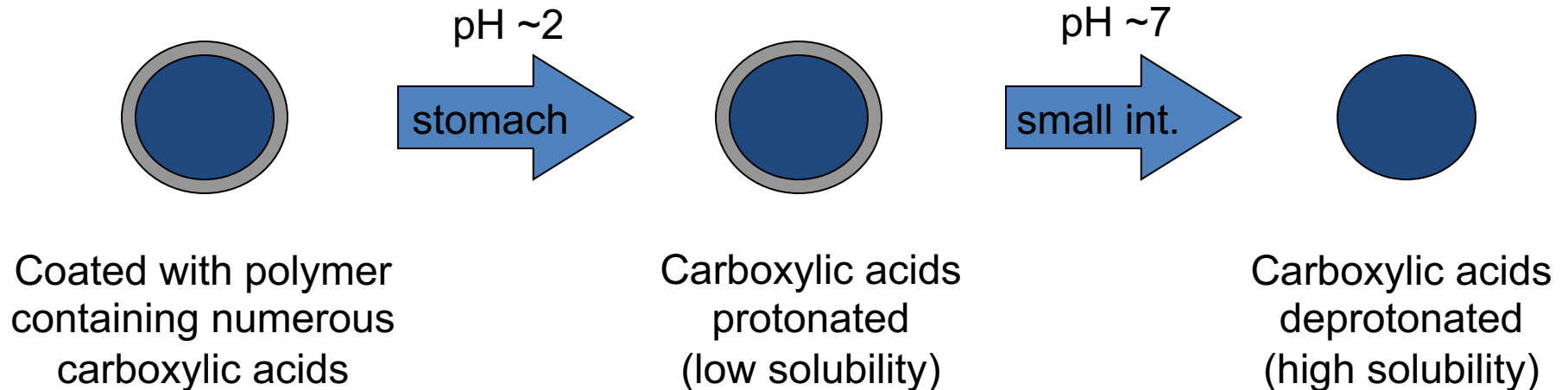
Osmotic tablets

- ***Driven by osmotic pressure***



What is an Enteric Coating?

A thin film applied to the dosage form to minimize drug dissolution in stomach



Subcutaneous

- Advantages
 - Patient self-administration
 - Slow, complete absorption
 - By-pass FPM
- Disadvantages
 - Invasive
 - Irritation, inflammation
 - Maximum dose volume - 2mL

Bydureon

- Glucagon like peptide
- Effective for type-2 diabetes
- Need injection twice daily
- PLGA microparticle formulation: Once a week:
SC administration



Intramuscular

- Advantages
 - Patient can administer the drug himself
 - Larger volume than subcutaneous
 - By-pass first pass metabolism
- Disadvantages
 - Invasive – patient discomfort
 - Irritation, inflammation
 - May require some training



Biodegradable Polymer (PLGA), 40 μ m PLGA particles

Drug: luteinizing hormone-releasing hormone (LHRH) analog

Name: Decapeptyl SR

Use: Advanced prostate cancer

Dose: ~3 months

Intramuscular Injection

Risperdal® Consta



- First long-acting atypical antipsychotic
- IM injection every two weeks versus daily treatment

Schizophrenia: Mental Disorder that results in delusions and hallucinations

Risperdal: Antipsychotic medication

Bio-erodible microspheres: Injected every two weeks instead of daily administration

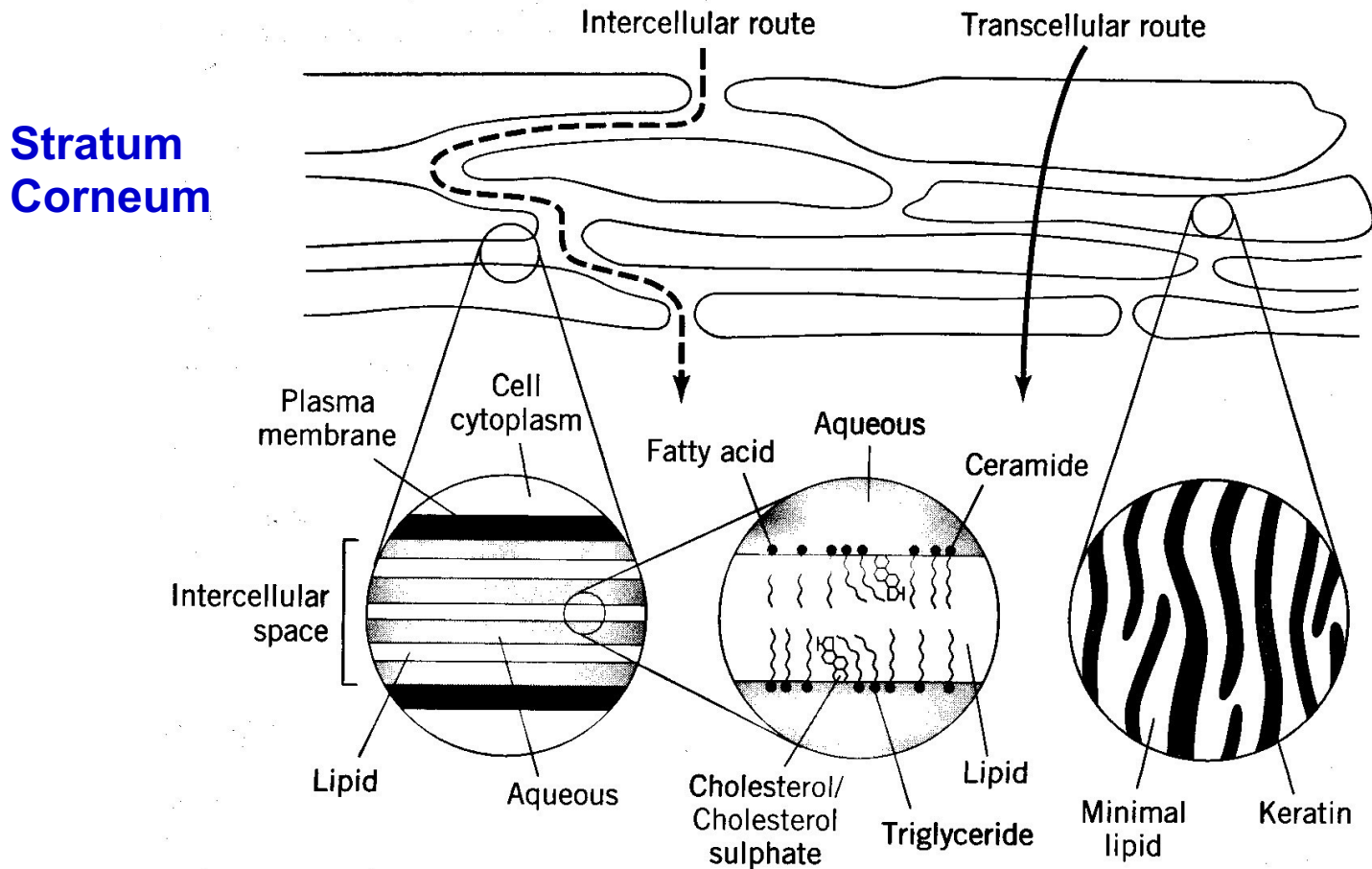
>5 million patients have taken this.

Transdermal

- Advantages
 - Local effect
 - Ease of administration
- Disadvantages
 - Low absorption for some drugs
 - May cause allergic reactions
- Requirement
 - $MW < 1,000 \text{ Da}$



Trans-dermal delivery: Possible micro-routes



BLOCKS OF “DEAD” KERATINOCYTES EMBEDDED IN A ‘MORTAR’ OF LIPIDS THESE LIPIDS ARE ARRANGED IN INTERLOCKING STRUCTURE, “BRICK AND MORTAR” STRUCTURE. THE LIPID “MORTAR” IS ARRANGED IN BILAYERS WHICH FORMS SHEETS AND CONSTITUTES THE MAJOR BARRIER

Transdermal drug delivery systems (TDDS)

- Advantages:
 - Reliable blood levels (steady levels can be achieved)
 - No first-pass metabolism
 - Very non-invasive → high patient compliance, provides simple localization → easy to apply and remove
- Disadvantages / limitations:
 - Skin is impermeable to most substances (main barrier: SC)
 - Lag time to reach steady state (generally several hours → skin loading and diffusion through SC)
 - Mostly useful for drugs with
 - High skin permeability (high partition coefficient in both water and oil) i.e. need to have some hydrophilicity and good lipophilicity
 - Low molecular weights

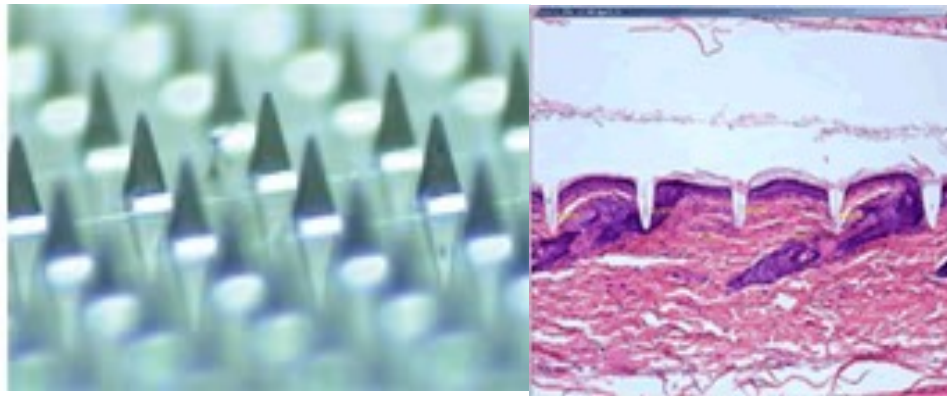
SKIN PERMEABILITY CAN BE ENHANCED USING PHYSICAL AND CHEMICAL MECHANISMS

Transdermal drug delivery systems (TDDS)

- Matrix type systems
 - FemPatch (Estradiol) → Cygnus
 - Vivelle (Estradiol) → Nova
 - Climara (Estradiol) → 3M
- Reservoir type systems:
 - Transderm-nitro (Nitroglycerin) → Angina treatment (per day)
 - ALZA-CIBA
 - Microporous membrane (EVAc)
 - Delivers 0.5 mg/cm²/day
 - Transderm-scop (Scopolamine) → motion sickness (per 3 days)
 - Microporous polypropylene membrane
 - Delivers 0.5 mg/day
 - Nicoderm CQ (Nicotine)

Example from Industry: 3M Company

- **Microstructured Transdermal System: MTS**
 - Microneedle system
 - Drug-in-adhesive technology platform



(a) 3M microneedle system and (b) histological section of microneedles in guinea pig skin

Micro-fabricated micro-needles for transdermal drug delivery

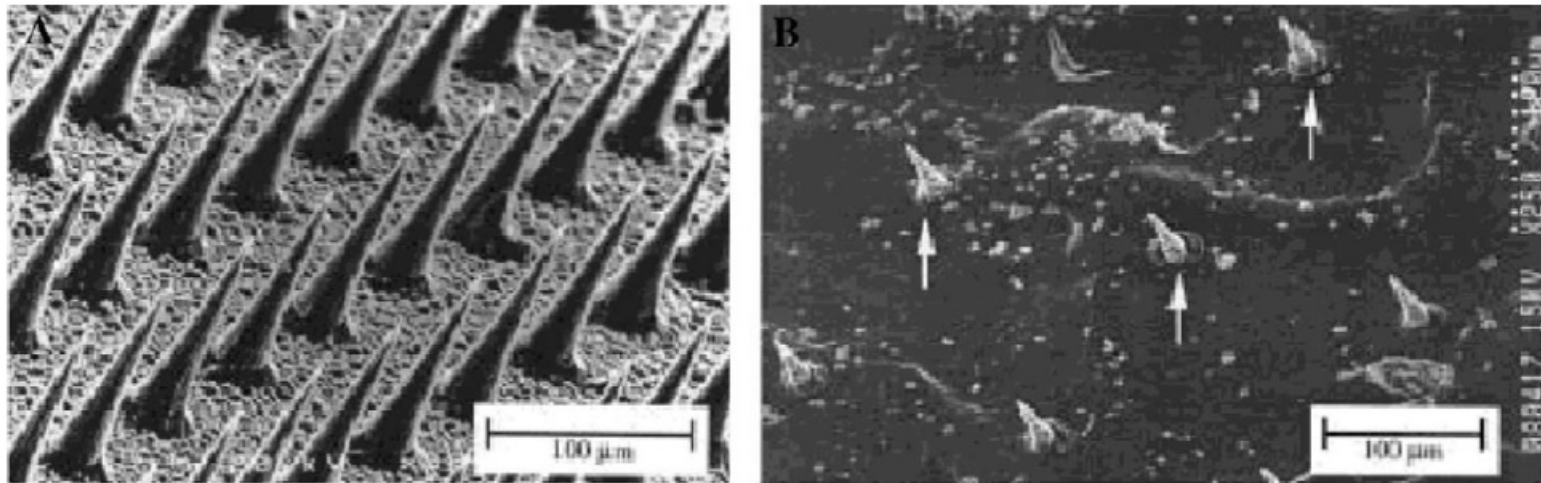


Fig. 1. (A) Scanning electron micrograph of microneedles made by reactive ion etching technique. (B) Microneedle tips inserted across epidermis. The underside of the epidermis is shown, indicating that the microneedles penetrated across the tissue and that the tips were not damaged. Arrows indicate some of the microneedle tips. Reproduced with permission from Ref. [14].

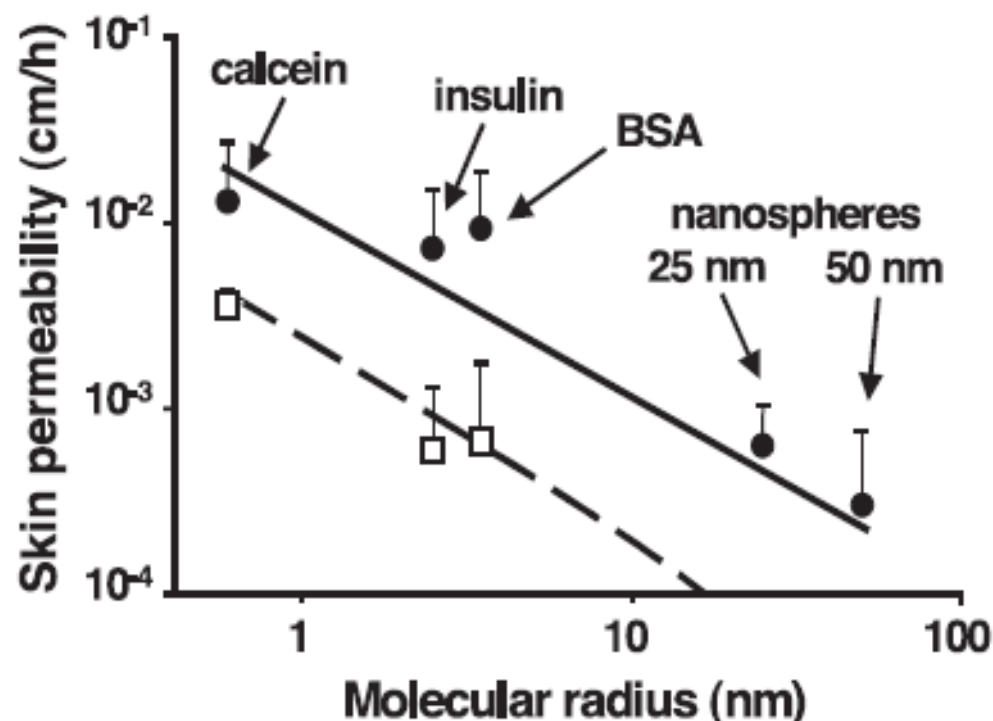


Fig. 4. Skin permeability to molecules and particles of different sizes after treatment with microneedles. The permeability of human cadaver epidermis was increased by orders of magnitude with a 400-needle array (Fig. 1A) inserted (□) and after the array was removed (●) for calcein, insulin, BSA, and latex nanospheres of 25 nm and 50 nm radius. Permeability to nanospheres with needles inserted was below the detection limit, on the order of 10⁻⁴ cm/h. In the absence of microneedles, permeability to all compounds was below their detection limits, on the order of 10⁻⁶ to 10⁻⁴ cm/h (data not shown). Mean values ± SEM are shown for at least six replicates. Predictions are shown for needles inserted (dashed line) and needles removed (solid line) by using a model requiring no adjustable parameters (Eq. 1 coupled with the Stokes–Einstein equation to interrelate molecular radius and diffusivity).

Hollow microneedles

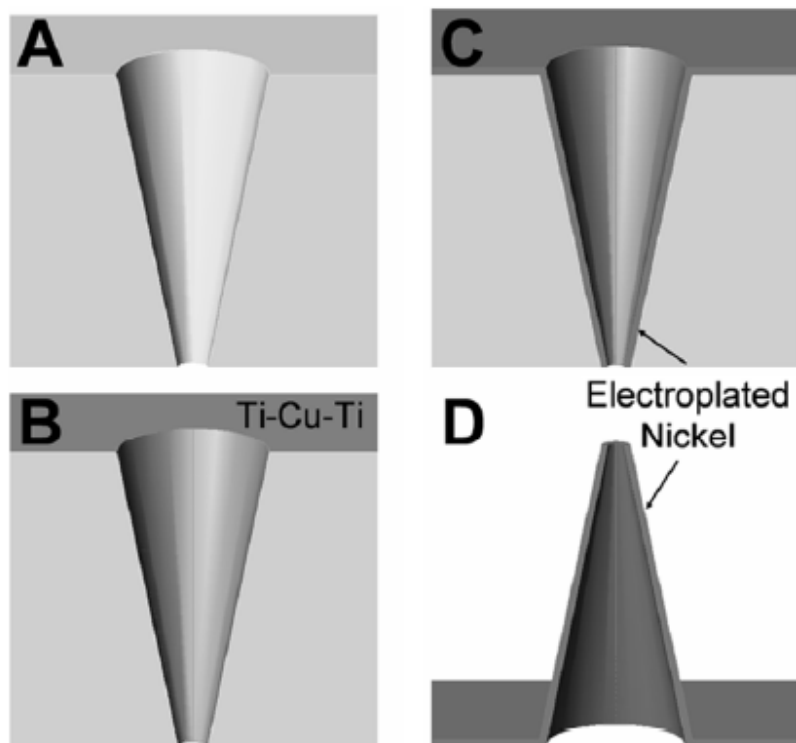


Fig. 1. Schematic sequence of the microneedle fabrication process using a polymer mold: (a) tapered holes are laser drilled through a polymer substrate, (b) a conductive seed layer is deposited on the top and sidewalls of the mold, (c) metal is electroplated onto the seed layer and (d) the mold is selectively wet etched to release the metal microneedles.

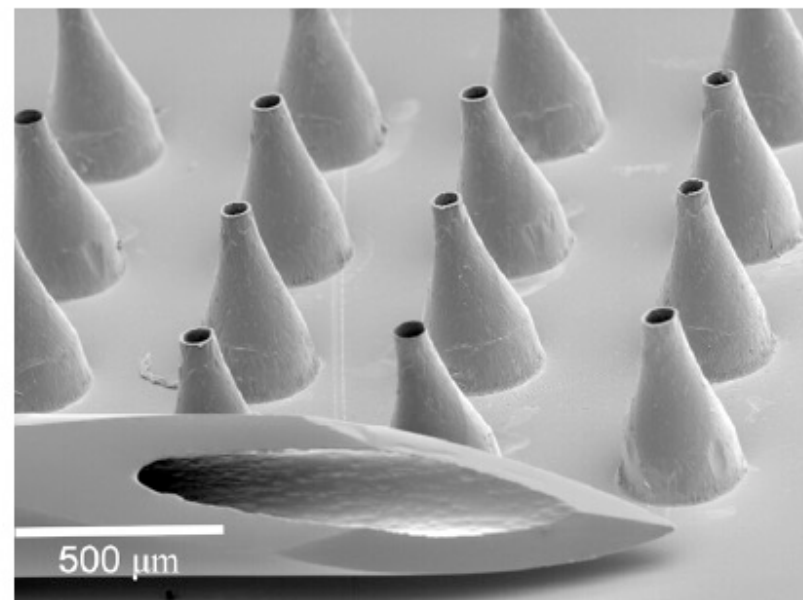


Fig. 5. An array of hollow, metal microneedles shown next to a 27-gauge hypodermic needle. The microneedles taper from a 300- μm base to a tip diameter of 75 μm over a 500- μm length and are arranged in a 4×4 array (i.e., 16 needles). Arrays of this geometry were used for insulin delivery experiments. Image from scanning electron microscopy.

Delivering insulin across skin using hollow microneedles

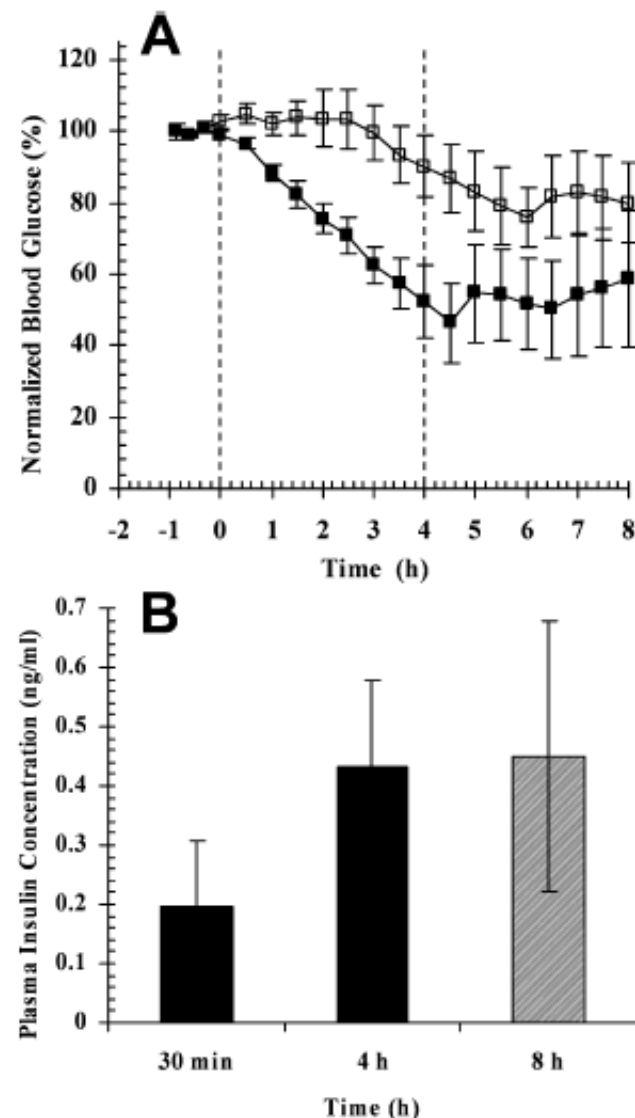
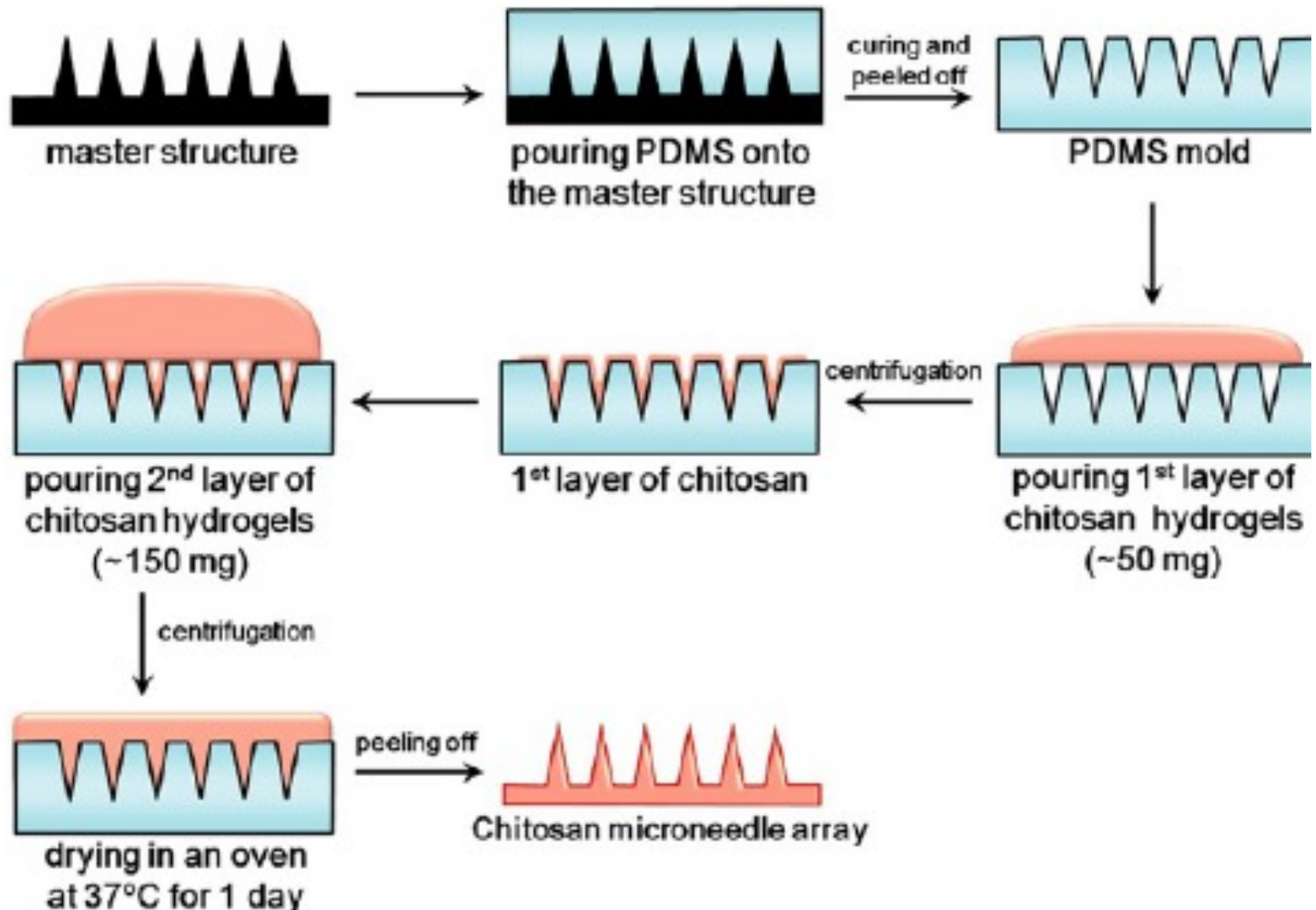


Fig. 6. Effects of transdermal insulin delivery using microneedles on blood glucose level and plasma insulin concentration in diabetic, hairless rats. (a) Blood glucose level before, during, and after transdermal insulin delivery either using microneedles inserted into the skin (■) ($n = 5$) or across intact skin (□) ($n = 4$). Glucose levels are normalized relative to the average blood glucose level during the 1.5-h period before treatment, 465 mg/dl. The dashed lines indicate the beginning and end of the 4-h insulin delivery period. (b) Concentration of human insulin in rat plasma during (0.5 and 4 h, black bars) and after (8 h, gray bar) transdermal delivery using microneedles. Average $\pm$ SEM is shown for $n \geq 3$ replicates.

Biodegradable microneedles



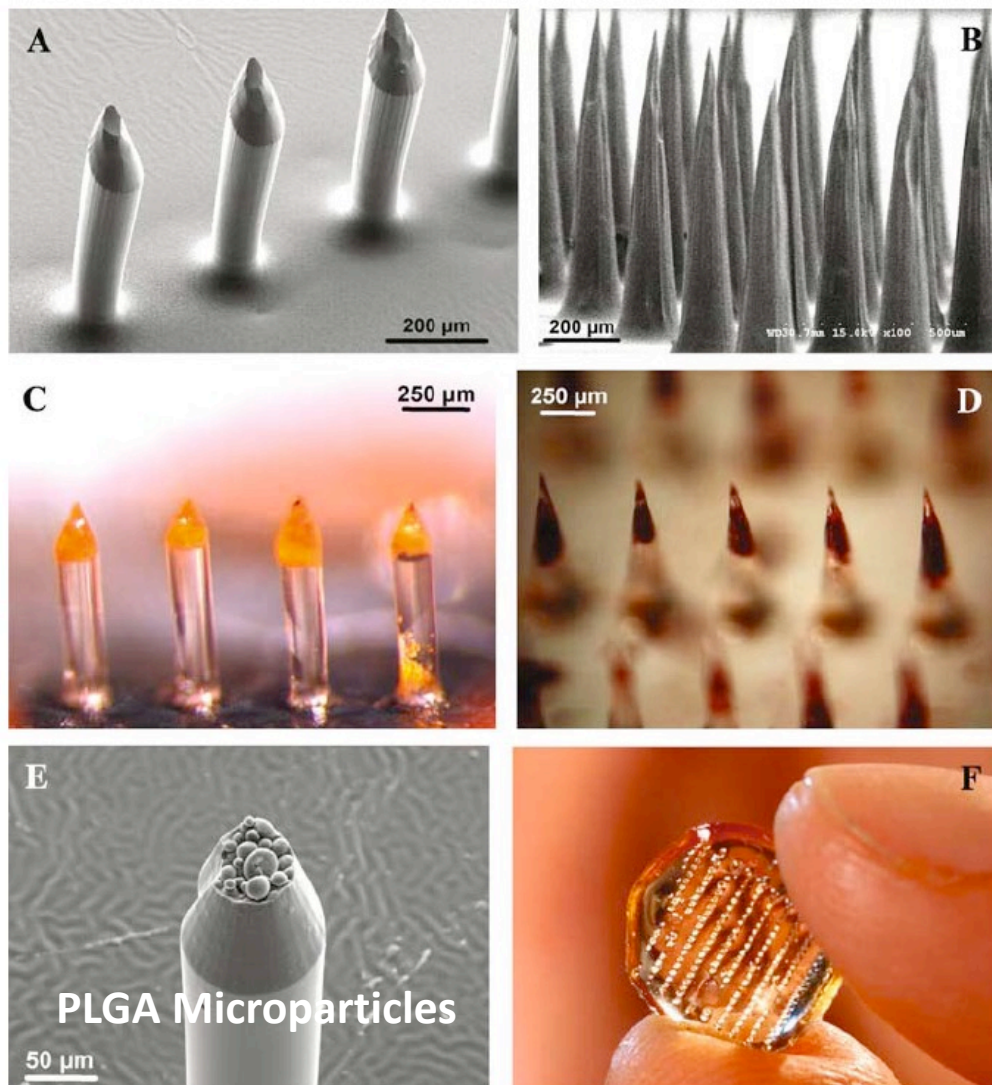


Fig. 3. Microscopy images of microneedles. A section of an array of (A) bevel-tip microneedles and (B) tapered-cone microneedles used as master structures (imaged by SEM). Making a mold and using it to prepare polymer microneedles as described in Fig. 2 yielded (C) bevel-tip and (D) tapered-cone microneedles made of PLGA and encapsulating calcein within their tips (imaged by fluorescence and bright-field microscopy, respectively). Using a double-encapsulation method produced microneedles that encapsulate microparticles that, in turn, encapsulate calcein. (E) Cutting off the tip of a PLGA microneedle reveals the PLA microparticles within (imaged by SEM). (F) A complete 20×10 array of PLGA microneedles is shown (imaged by flash photography).

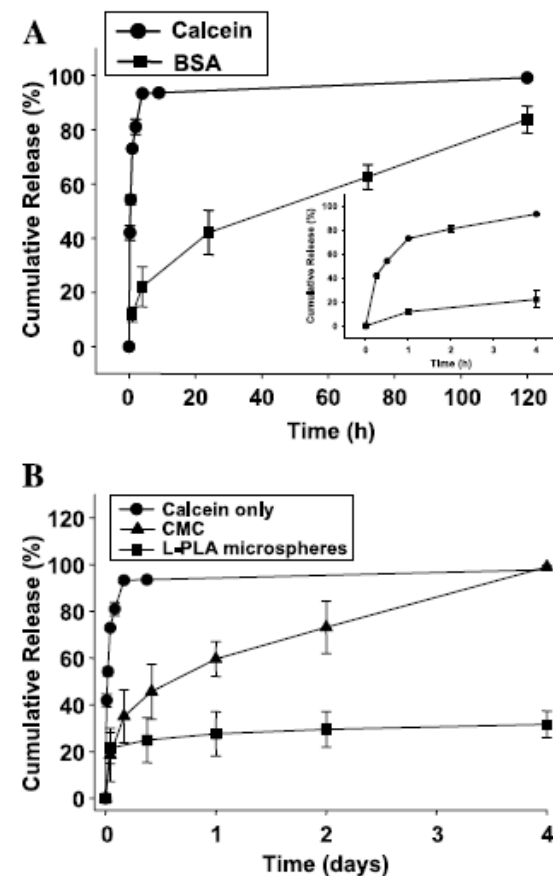


Fig. 4. Cumulative release of model compounds from PLGA microneedles prepared using different formulations to control release kinetics *in vitro*. (A) Release of calcein (●) and BSA (■) encapsulated within microneedles. The inset has an expanded time axis to better show release over the initial 4 h. (B) Release of calcein from microneedles that encapsulated (●) calcein only, (▲) CMC microparticles containing calcein, or (■) PLA microparticles containing calcein. Data are presented as average $\pm$ SEM ($n = 3-5$).

Inhalers

- Advantages

- By-pass FPM
- Good for delivery to lungs
- Patient compliant

- Disadvantages

- Systemic absorption only occurs for molecules that reach deep lungs
- Solids and liquids can be absorbed in systemic circulation only if size is below 100 nm



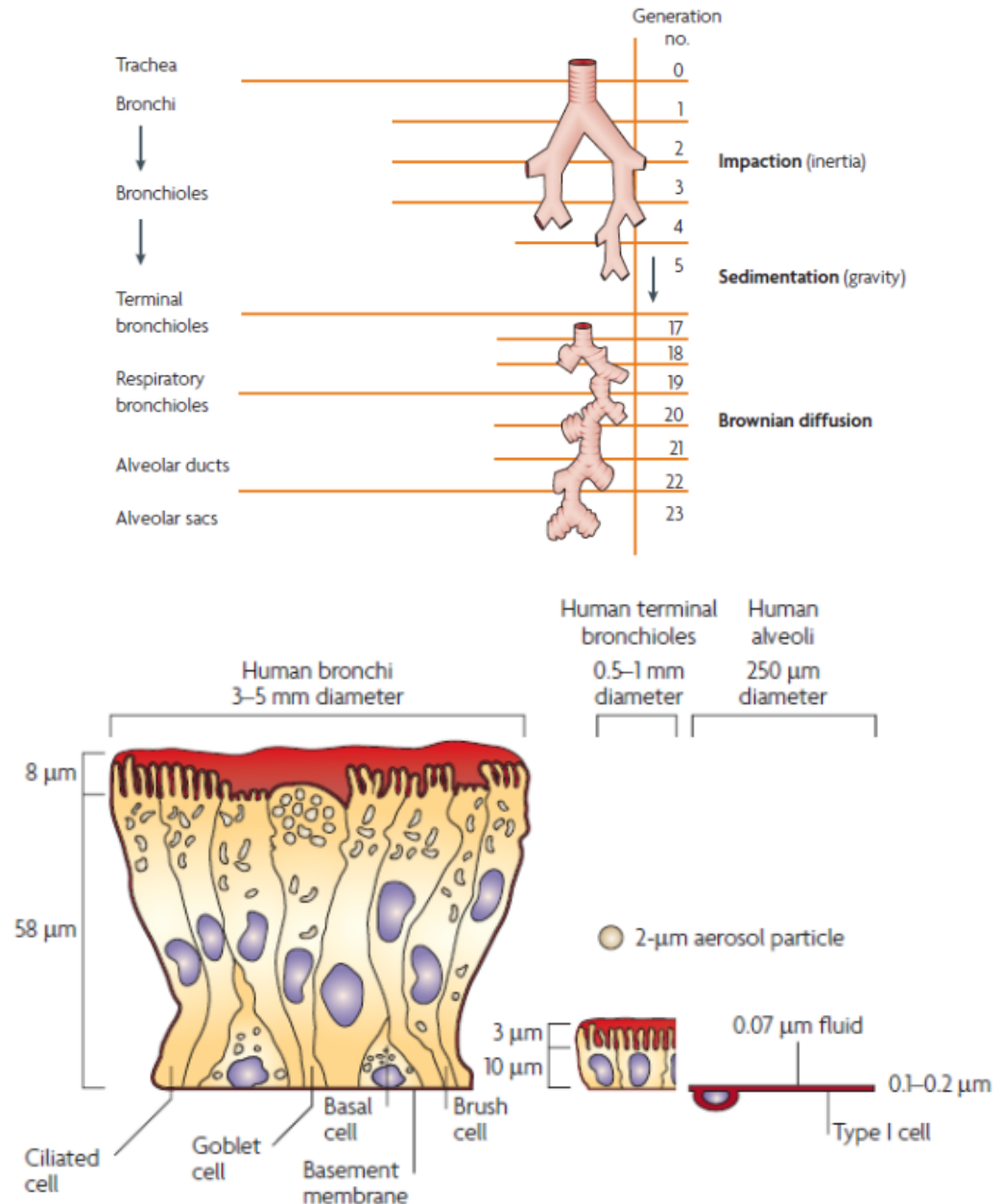
Inhalation based drug delivery

Table 1 | **Examples of inhaled medicines**

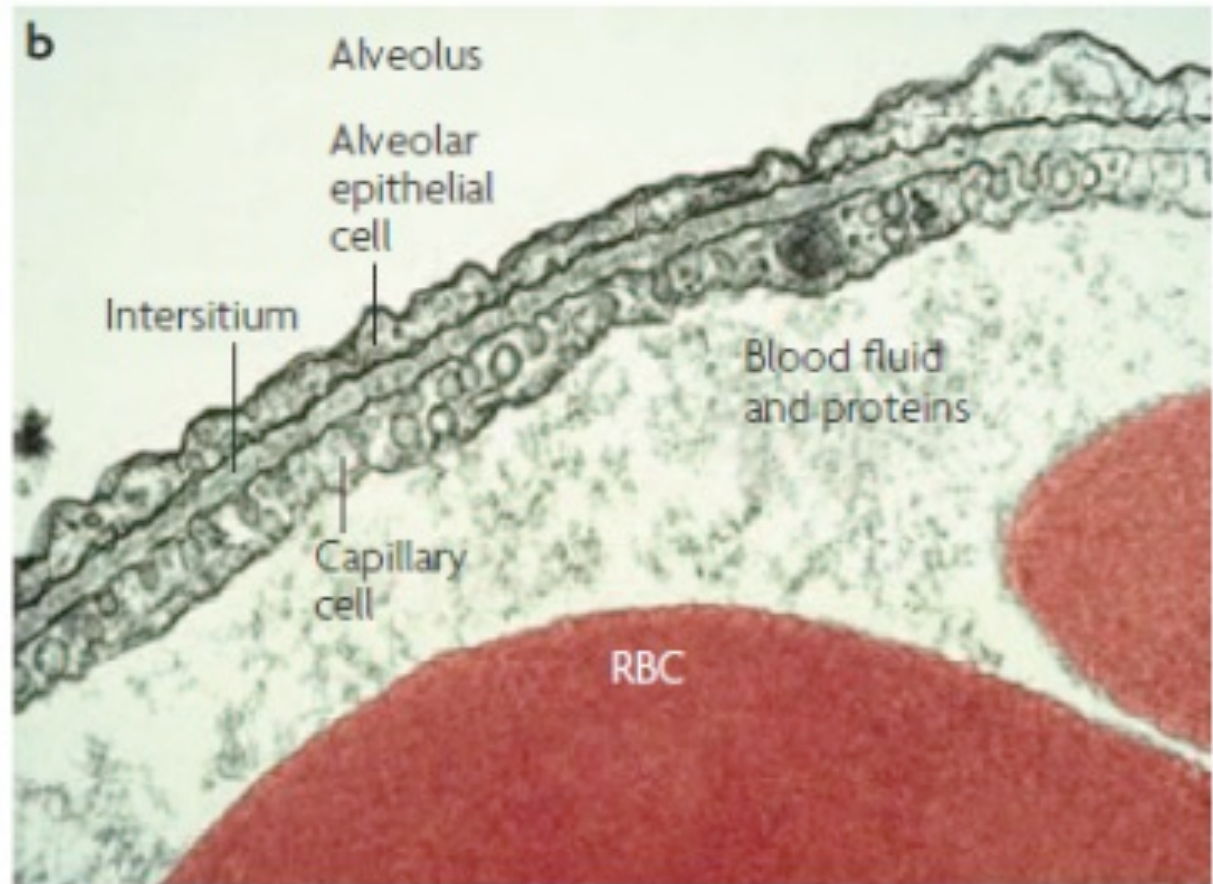
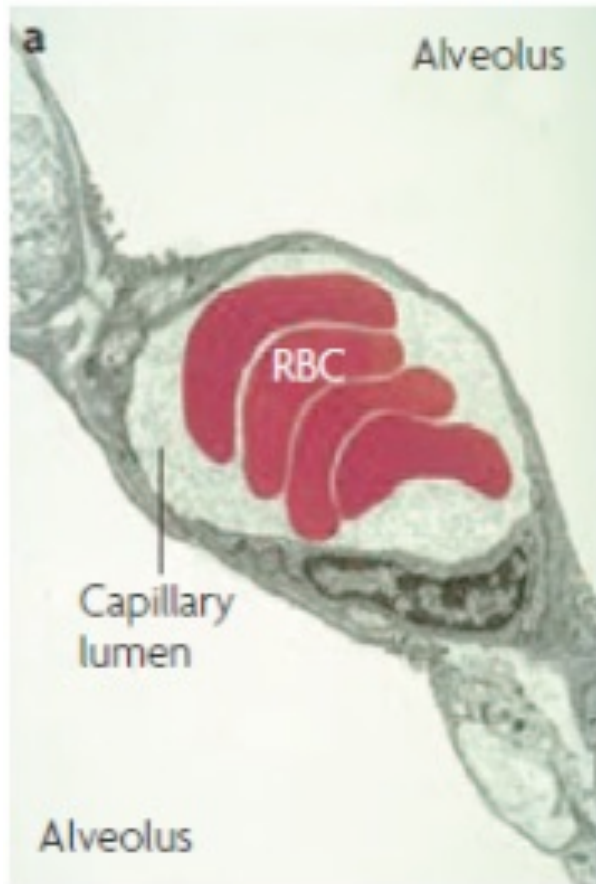
Year	Formulation/device	Molecule(s)	Disease
1500 BC	Egyptians used 'vapors'	?	?
1662 AD	Bennet's inhalation treatment	?	Tuberculosis
1802	Potter's cigarettes	?	Asthma
1860	Sales-Giron's portable nebulizer	?	?
1925	Aqueous/nebulizer	Insulin	Diabetes
1945	Aqueous/nebulizer	Penicillin	Lung infections
1951	Aqueous/nebulizer	Isoprenaline	Asthma
1955	Aqueous/nebulizer	Hydrocortisone	Asthma
1956	First metered-dose inhaler (MDI) (freon)	Albuterol	Asthma
1960	First dry powder inhaler (DPI)	Norsadrenaline	Asthma
1988	First multidose DPI	Terbutaline	Asthma
1996	First protein aqueous/nebulizer	DNase	Cystic fibrosis
1998	First antibiotic aqueous/nebulizer	Tobramycin	Cystic fibrosis
1997	First hydrofluoralkane MDI	Albuterol	Asthma
2006	First protein DPI	Insulin	Diabetes

Inhalation

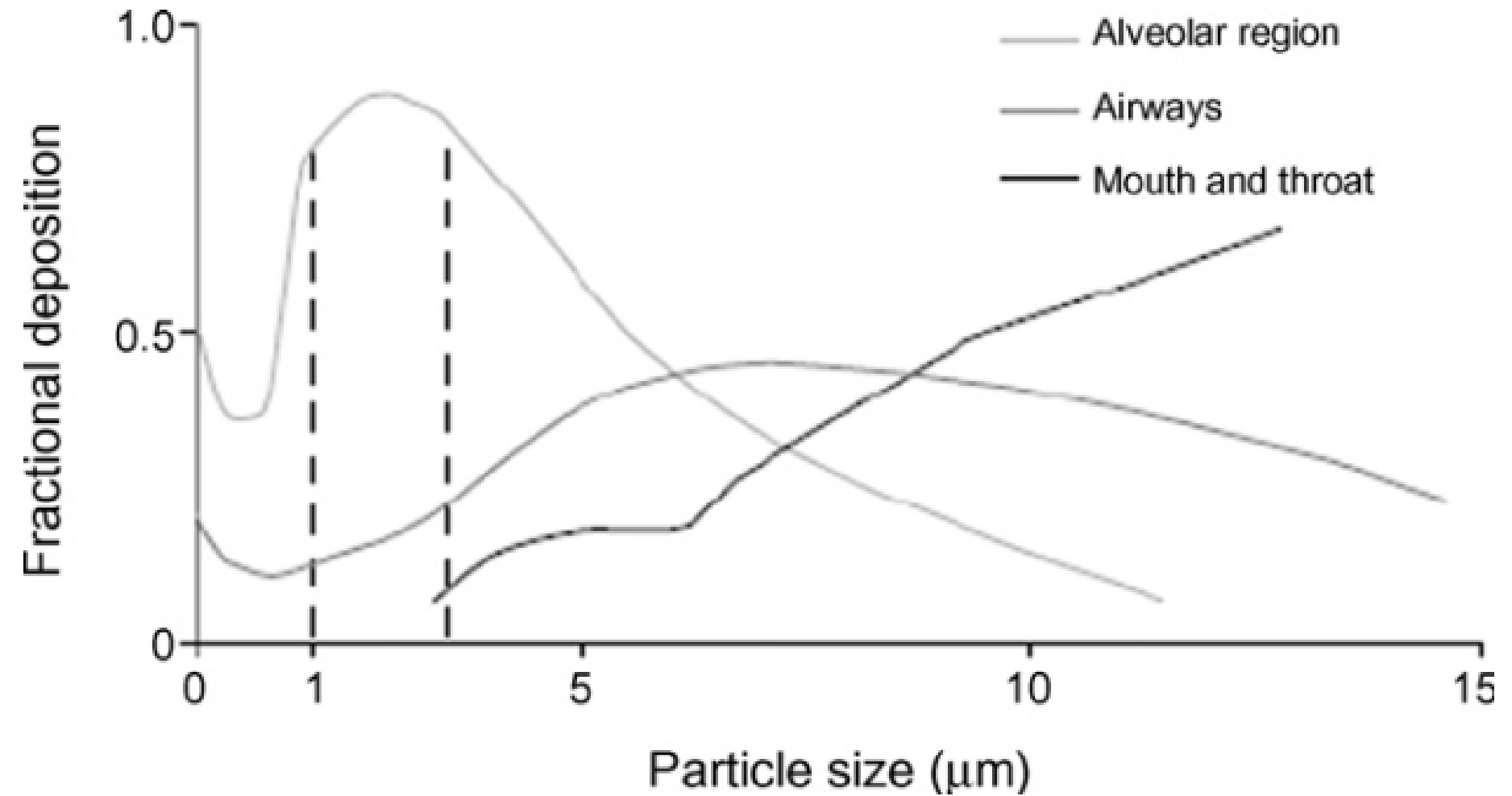
- The airways branch roughly 16–17 times before alveolar sacs are encountered
- About 500 million alveoli in human lungs
- Each alveoli is patrolled by 10-15 alveolar macrophages



Lung-capillary permeability

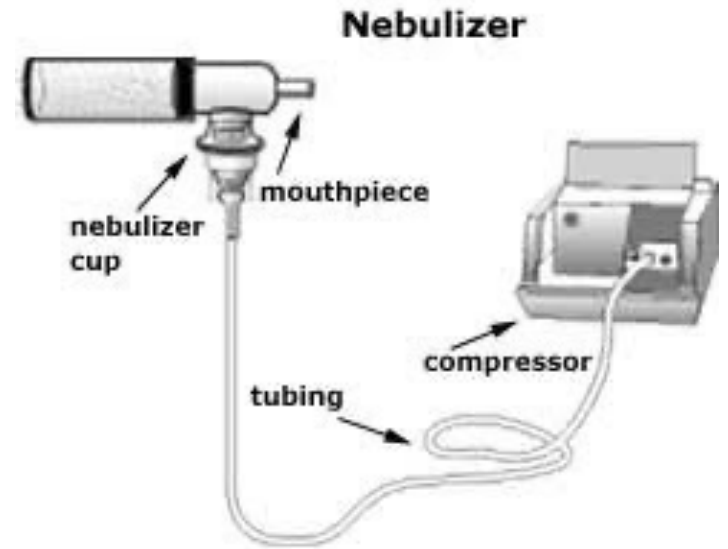


Inhalation



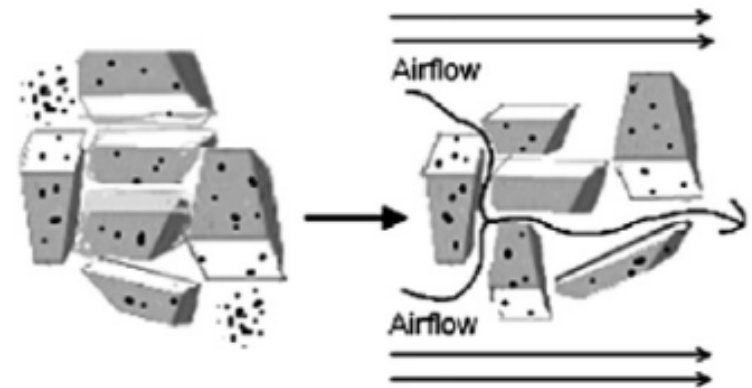
Nebulizers

- Generates mist that are inhaled in the lungs
- Use compressed air and ultrasonic power to break solutions into aerosol droplets
- Works well in a hospital setting but not very easy and convenient to use otherwise



Dry powder inhalation

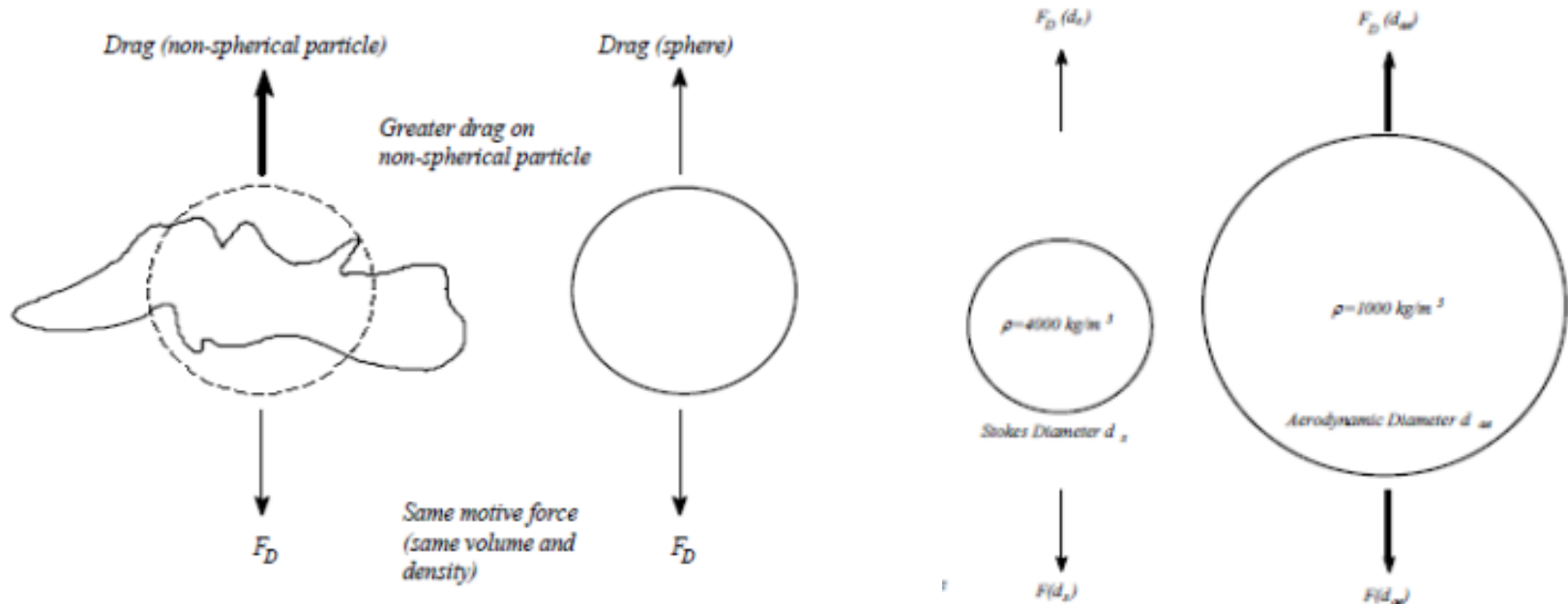
- Patient compliant/ease of use
- Drug stability is increased
- Powder particles should be around 1-5 μm for deep lung delivery
- Excipients are used to carry drug adsorbed on the surface to deep lungs
- Sugars such as lactose are commonly used
- Doesn't result in slow release as all the drug is administered immediately
- Particles of can also be used for this application and will result in depot at lung alveolar space



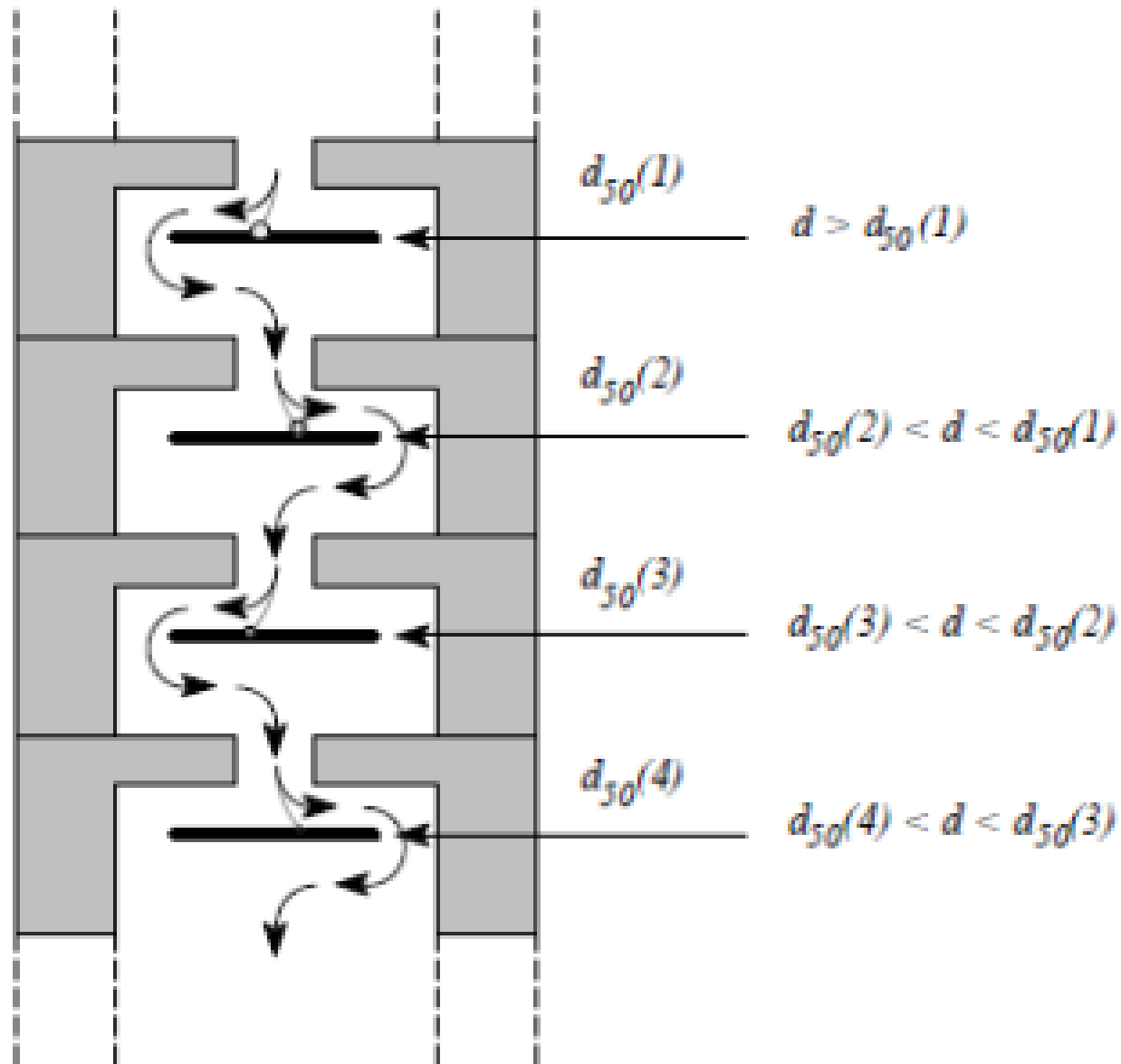
Inhalable Carriers: Aerodynamic Diameter

- Particle size for **deep lung delivery** via pulmonary route: 1-5 μm in **aerodynamic diameter**
- Aerodynamic Diameter: The diameter of sphere of density 1000 kg/m^3 with the same settling velocity as particle of interest
- Standardizes for shape (sphere) and density (water droplet)
- Measured by cascade impactor

Drag force for different shape of particle



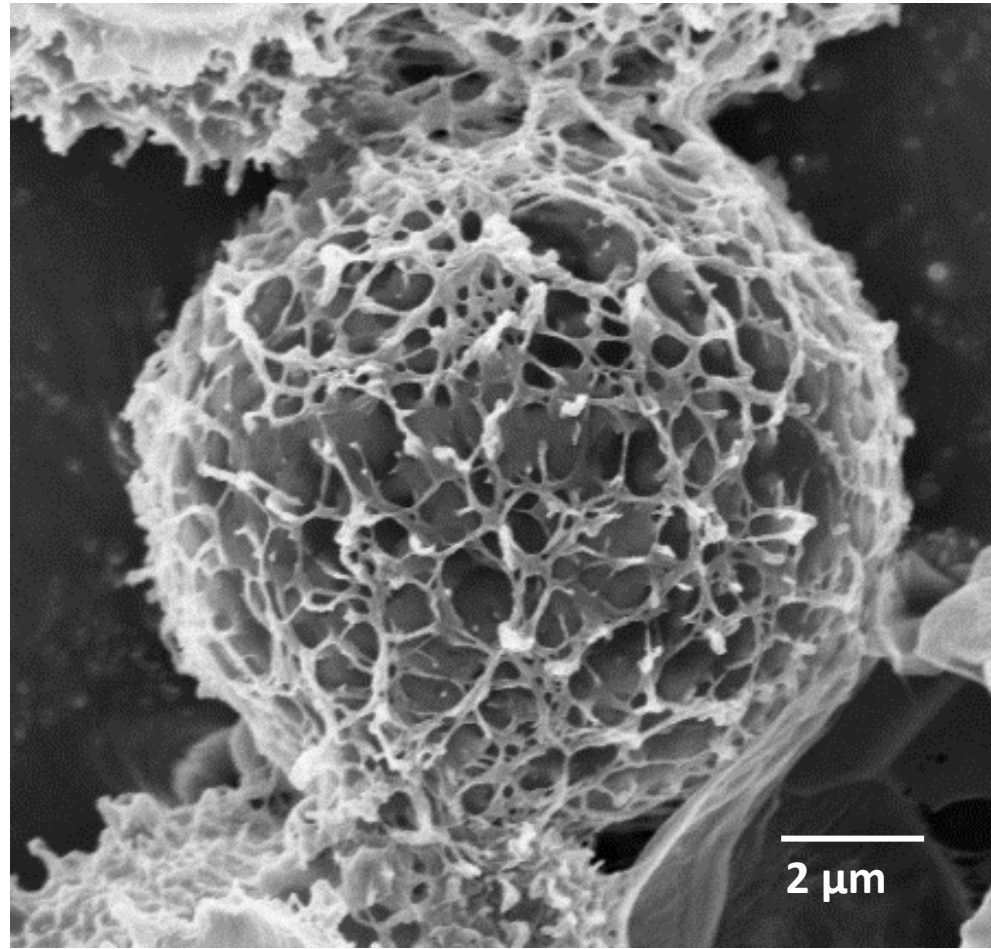
Cascade impactors



Inhalable Carriers: Porous Particles

- Particle size for **macrophage uptake**: 1-5 μm
- Porosity achieved by including porogens or effervescent molecules during synthesis
- **Porous & larger** particles
 - 8-10 μm in physical diameter
 - Porosity reduces particle density to match 2-5 μm d_{aer}
 - Cannot be phagocytosed by macrophages
 - Larger size: less aggregation and better aerosolization

$$D_{\text{aero}} \propto D_p^{1/2}$$



Agarwal et al. Nature Biomedical Engineering 2, 841 (2018).

Relenza: Anti-flu medication



- Blocks release of virus from cell surface
- Not much in use these days as replaced by other drugs

Rectal

- Advantages
 - By-pass first pass metabolism
 - Useful for children as might be difficult to find blood vessels
- Disadvantages
 - Absorption depends on disease state
 - Degradation by bacterial flora
 - Uncomfortable
- Traditional delivery system/devices
 - Suppository
 - a small, round or cone-shaped object that dissolves rapidly
 - Enema



Not heavy research is being done here due to low patient compliance

Example from Industry: Valeant Pharmaceuticals

- Diastat AcuDial: diazepam rectal gel for prolonged seizures
- High need for children and an administration route that parents can easily administer
 - Even in hospital setting, its hard to find the vein for babies and toddlers
- High lipid solubility
 - Rapidly diffuses into all high lipid regions
- Oral administration is slow to take effect and muscular administration causes necrosis in the region

Rectal administration

- Fecal microbiota transplant
- Clinically used for treatment of *Clostridium difficile* infection
 - Diarrhea
 - Fever
 - Abdominal pain
 - Vancomycin unable to clear it in many cases
- Stool from close family member is administered using enema
- Invasive, so not often used for other treatments

Intra-articular administration

- Direct injection to the joints
- Specific for joint disease

Zilretta

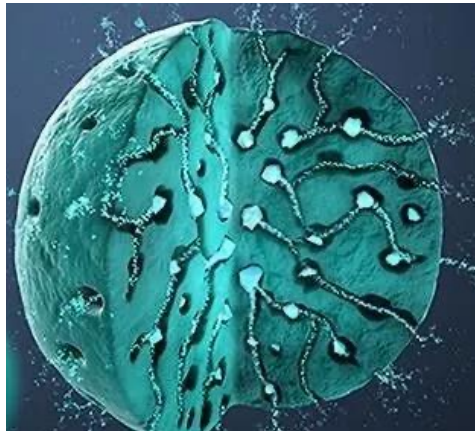


- 35-55 μm PLGA particles with around 500nm channels
- Active ingredient is triamcinolone acetonide, which is a common short acting corticosteroid
- Longer lasting pain relief for more than 12 weeks

\$500 per injection
\$111 million sales in 2023

<https://www.pharmaceutical-technology.com/comment/commentwith-fda-approval-secured-flexions-zilretta-poised-to-shake-up-knee-osteoarthritis-marketplace-5947301/>

<http://www.limu-chemicals.com/en/News/2018/0418/39.html>



Zilretta

- 32 mg of triamcinolone acetonide per injection in 5mL.
- Drug loading is 25%w/w so around 96mg of PLGA per injection
- Can only be given again atleast after 12 weeks of first administration

Intravenous (IV)

- **Advantages**
 - Drug 100% bioavailable
 - Rapid response
 - Total control of blood concentration
 - By-pass FPM
- **Disadvantages**
 - Invasive
 - Trained personnel
 - Possible toxicity due to incorrect dosing
 - Sterility
- **Traditional delivery system/devices**
 - Injection-bolus
 - IV bag - infusion

